WEEKLY TEST · MID QUIZ 1 · WED 28 OCT 2026 · RETAKE · FINAL SCORE

DSA + System Design

DSA #14 Longest Common Prefix · System Design Lesson 7 Cache — Sections 1–4 · 2026-10-28 · Closed-book

30 / 40 **75%** ✓ PASS

PASS TARGET

28 – 32

Scored 30 — inside target

☰ MULTIPLE CHOICE 20 / 20

DSA 9 / 9 · System Design 11 / 11 · 100%

✎ ESSAYS 10 / 20

E1 4 / 5 · E2 3 / 5 · E3 2 / 5 · E4 1 / 5

★ STRENGTHS

- All 20 MC correct.
- Cache choices correct.
- Cache-miss flow correct.

▲ FIX NEXT

- Slicing: `flower[:4] = flow`
- Loop bound: `len(prefix)`, not `len(strs)`
- Essays: answer every part a and b.
- Fix: match the failure mode.

Part A — DSA Multiple Choice

Longest Common Prefix · Pairwise shrink method · Q1–Q9 · 9 / 9

Q1 ✓ 1 / 1

How does the pairwise method start?

A. Take the first string as the temporary prefix ✓

Your answer: A = Correct answer: A

Q2 ✓ 1 / 1

Arrange the pseudocode steps in the correct order.

1. Take the first string as the prefix **2.** For each next string, compare the prefix with that string character by character from the start **3.** Keep only the matching characters, and cut the prefix to that length **4.** If no characters match, return "" at once **5.** After the last string, return the prefix

C. 1-2-3-4-5 ✓

Your answer: C = Correct answer: C

Q3 ✓ 1 / 1

What does shrink mean?

A. The prefix only shrinks or stays the same, it never grows longer ✓

Your answer: A = Correct answer: A

Q4 ✓ 1 / 1

When do you return "" at once?

A. When the prefix becomes "" after a string ✓

Your answer: A = Correct answer: A

Q5 ✓ 1 / 1

In Python, what does prefix[:i] mean?

A. Take the first i characters of the prefix, not including the character at position i ✓

Your answer: A = Correct answer: A

Q6 ✓ 1 / 1

What does "flower"[:4] give, and why can it be used to shrink the prefix?

A. "flow", because it keeps the first 4 characters and cuts the rest ✓

Your answer: A = Correct answer: A

Q7 ✓ 1 / 1

In JavaScript, which command is equivalent to Python prefix[:i]?

A. prefix.slice(0, i) ✓

Your answer: A = Correct answer: A

Q8 ✓ 1 / 1

What does this code do, and why is it safe? if prefix == "": return ""

A. Stop at once and return "" when there is no common prefix left, with no need to compare further, because a common prefix cannot grow back ✓

Your answer: A = Correct answer: A

Q9 ✓ 1 / 1

What are the key idea and the complexity of the pairwise method?

A. Time O(n*m), equal to the total number of input characters, Space O(1) extra, because each character is compared and the prefix only shrinks ✓

Your answer: A = Correct answer: A

Part A — System Design Multiple Choice

Cache — Sections 1–4 · Q10–Q20 · 11 / 11

Q10 ✓ 1 / 1

What is the main idea of a cache?

C. Store a result already computed or fetched, so the next request does not repeat the expensive work ✓

Your answer: C = Correct answer: C

Q11 ✓ 1 / 1

A Redis cache hit takes about 1 ms. A database query often takes 20-50 ms. What does this show?

B. Cache is about 20-50 times faster for a hit ✓

Your answer: B = Correct answer: B

Q12 ✓ 1 / 1

What is true about client-side cache?

D. It is the fastest cache, but hard to control removing old data ✓

Your answer: D = Correct answer: D

Q13 ✓ 1 / 1

A distributed cache has 3 nodes. Each node holds one part of the data. One node goes down. What happens?

A. Only that node's part is lost ✓

Your answer: A = Correct answer: A

Q14 ✓ 1 / 1

One shared Redis cache stores user sessions. All application servers read the same cache. What can happen next?

C. Any server can handle the next request ✓

Your answer: C = Correct answer: C

Q15 ✓ 1 / 1

With query-level caching, one cell in a table changes. What must happen?

D. Delete all cached queries that contain that cell ✓

Your answer: D = Correct answer: D

Q16 ✓ 1 / 1

Which group fits object-level caching?

A. User session, rendered web page, activity stream ✓

Your answer: A = Correct answer: A

Q17 ✓ 1 / 1

In cache-aside, what happens on a cache miss?

D. Load the entry from the database, write it to the cache, then return it ✓

Your answer: D = Correct answer: D

Q18 ✓ 1 / 1

What is the flow of write-through?

C. Application writes to the cache, cache writes to the database at the same time, then it returns ✓

Your answer: C = Correct answer: C

Q19 ✓ 1 / 1

What is the main risk of write-behind?

C. Data can be lost if the cache crashes before the data reaches the database ✓

Your answer: C = Correct answer: C

Q20 ✓ 1 / 1

What is refresh-ahead?

B. The cache refreshes just-accessed entries by itself, before they expire ✓

Your answer: B = Correct answer: B

Multiple Choice — 20 / 20

DSA 9 / 9 · System Design 11 / 11 · Every answer correct

Part B — Essays

Full prompt · Your answer · Score · Grading reason · 10 / 20

Essay 1 — 4 / 5

System Design — Cache-aside for reads

PROMPT

A product page is read many times a day and the database is overloaded. The application controls reads and writes. Choose cache-aside or write-through for reads and say why. Give the cache-miss flow steps in order. A replacement cache node starts empty — what happens, and how can you reduce that problem?

YOUR ANSWER

It is a read-heavy way. I would like to use cache-aside for the read. The reason is data changes less.
Flow: request → check cache (miss) → request db → cache save db → return data. When a replacement cache node starts empty, it will take longer for the first time, after that it will be faster.

Grading reason: Choice and flow correct — missing the reduction method for the empty-node problem.

Essay 2 — 3 / 5

System Design — Write-through vs write-behind

PROMPT

A user places an order. The order must not be lost, and the user reads it right after placing it. Choose write-through or write-behind and state the trade-off. What fails if write-behind is used and the cache crashes before the data reaches the database? How do you fix it?

YOUR ANSWER

Write-through, as it might be a little late but consistency.
b) The data might be lost. Fix it by use distributed cache.

Grading reason: Choice and risk correct — the fix is incorrect: a distributed cache alone does not stop write-behind data loss.

Essay 3

2 / 5

DSA — Re-code #14 Longest Common Prefix

PROMPT

Re-code Longest Common Prefix in Python using the pairwise shrink method. Explain your thinking step by step in English. Trace your code on ["flower", "flow", "flight"]. State the time and space complexity.

YOUR ANSWER — CODE

```
class Solution:
    def longestCommonPrefix(self, strs):
        if not strs:
            return ""
        prefix = strs[0]
        for word in strs[1:]:
            i = 0
            while i < len(word) and i < len(strs) \
                    and prefix[i] == word[i]:
                i += 1
            prefix = prefix[:i]
            if prefix == "":
                return ""
        return prefix
```

YOUR ANSWER — PART B

Part b not answered. Scored 0 for part b — no redo the next day.

Grading reason: Structure correct — bug: **len(strs)** instead of **len(prefix)** in the loop bound. Part b missing, scored 0.

Essay 4

1 / 5

DSA — Explain prefix = prefix[:i]

PROMPT

Explain the line prefix = prefix[:i]. Give the example with "flower". Say why this line works for shrinking the prefix. Then explain what if prefix == "": return "" does, and why it is safe.

YOUR ANSWER

a) i = 4 → return "flo"
b) If prefix is empty, return empty.

Grading reason: "flower"[:4] is "flow", not "flo" — why slicing works is missing, and why the early return is safe is missing.

TOTAL 30 / 40 = 75% ✓ PASS

Multiple Choice 20 / 20 · Essays 10 / 20 · Pass target 28–32